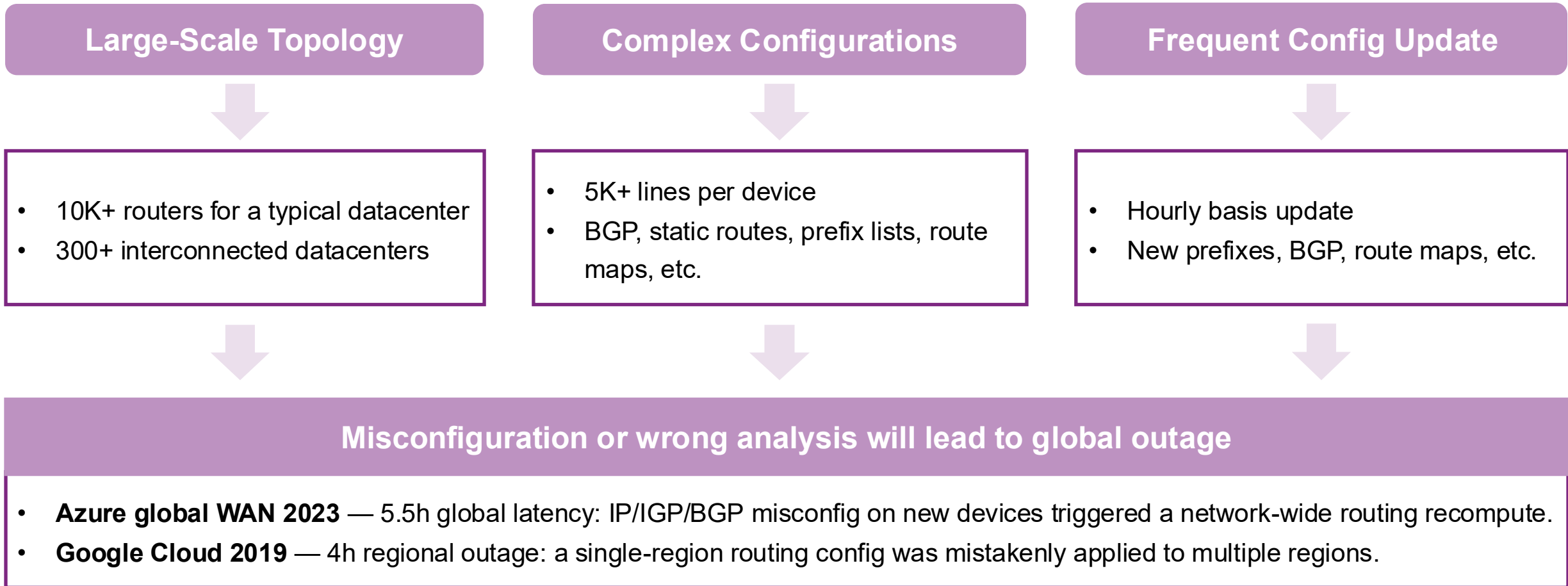


RoutingBench: Can Agentic Routing Analysis Scale to Production Datacenter Networks?

Wenlong Ding

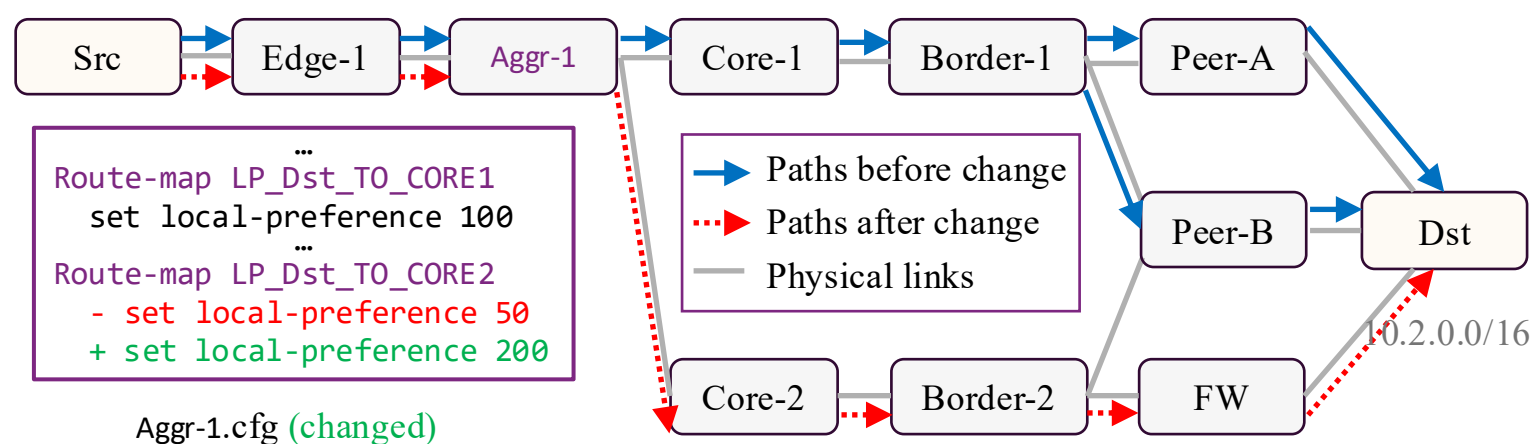
The Chinese University of Hong Kong

Analyzing Production Network at Scale Is Important



The NetOps Primitive: Path Analysis

10.1.0.0/16



```
! Static route for dst prefix 10.2.0.0/16
! Next hop is set to FW (ip: 192.168.1.1)
ip route 10.2.0.0 255.255.0.0 192.168.1.1
```

Border 2.cfg (Unchanged)

```
Route-map MED_Dst_from_Peers
set MED 100
```

Peer A/Peer B/FW.cfg (Unchanged)

Paths before change

- Aggr-1 → Core-1: $LP_1 = 100 > LP_2 = 50$
- Border-1 → Peer-A/B: $MED_A = MED_B = 100$

Paths after change

- Aggr-1 → Core-2: $LP_1 = 100 < LP_2 = 200$
- Border-2 → FW: $MED_B < Static_{FW}$

Challenge: Cross-protocol cross-device analysis

Agentic AI for Path Analysis: Potential and Challenge

LLMs are pretrained with network-protocol knowledge and can reason about protocol interactions.

Agentic technologies (e.g., skills with tools) enable LLMs to handle long contexts.



Agentic AI has potential for path analysis over topology and configs at scale.

The key aspect we study:

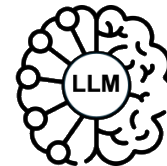
- ✗ Whether LLMs are literate in network configs
- ✓ How an agent efficiently discovers **decisive** config evidence under context pressure and long reasoning chains at scale.



```
Route-map LP_Dst_TO_CORE1  
set local-preference 100
```

```
Route-map MED_Dst_from_Peers  
set MED 100
```

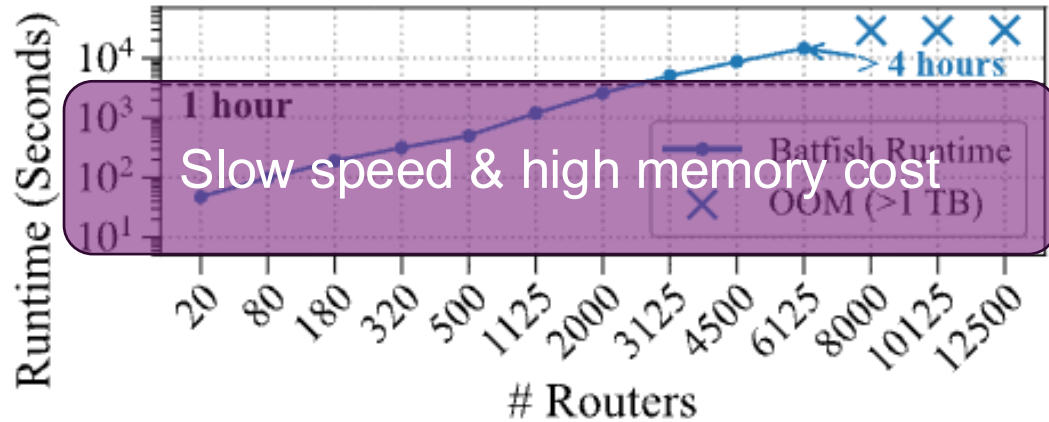
...



- 5K+ lines per router
- O(10K) routers per DC
- 300+ DCs

Reason over the collected config evidence to infer paths.

Scalability Bottleneck for Traditional Symbolic Tools



VS

Production Network Complexity

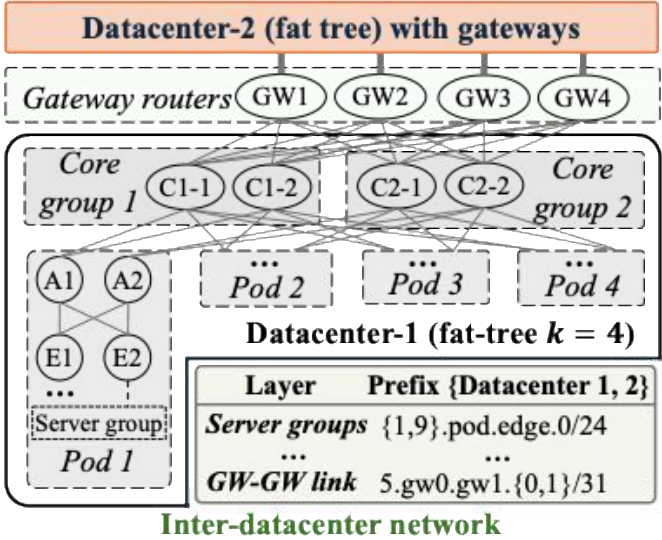
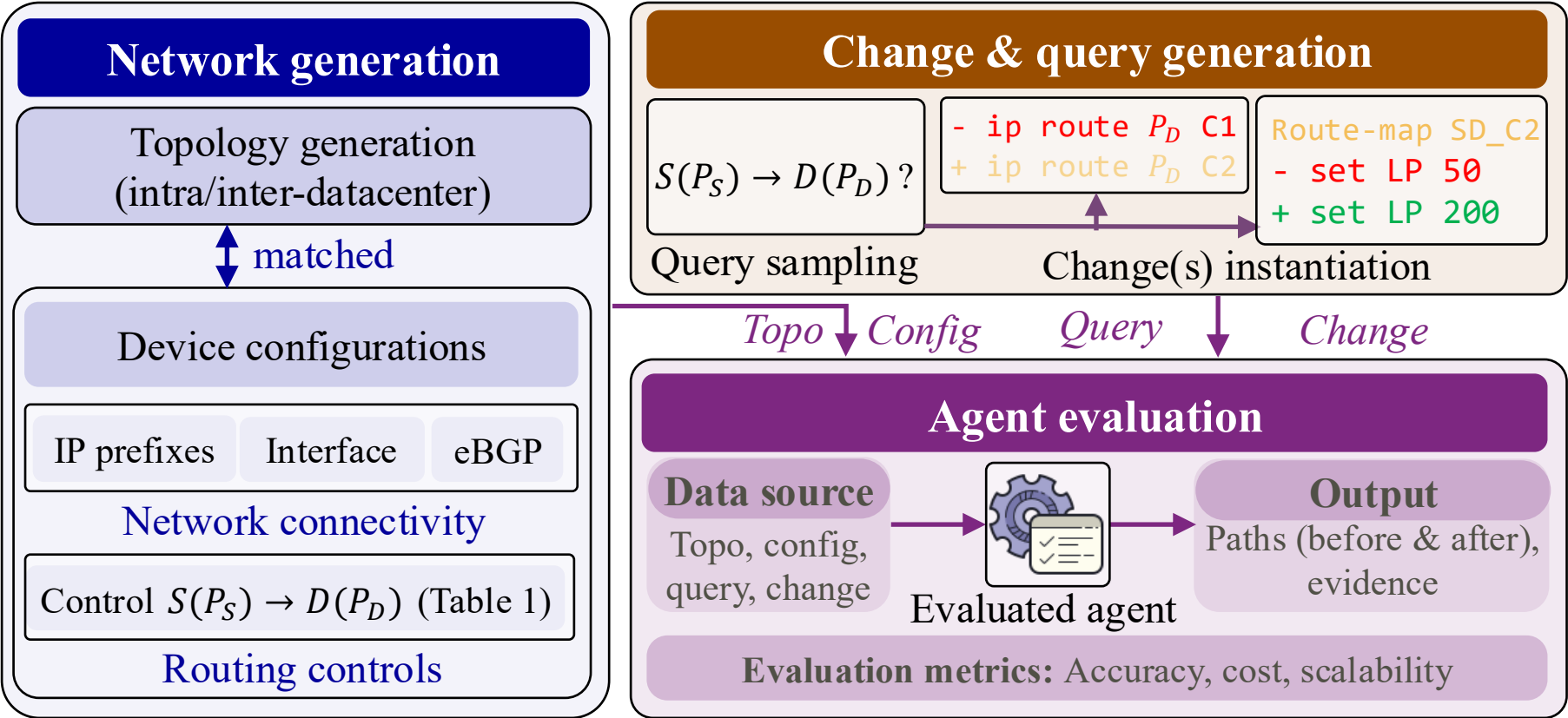
Hourly config changes
Strict timeliness & massive scale

- 10K+ routers per datacenter
- 300+ datacenters in total

Batfish on a 1 TB-RAM server can't even reach single-datacenter scale

RoutingBench: Challenging Agentic AI on Path Analysis at Scale

An automatic test generator at arbitrary scale — not a hand-picked test set.



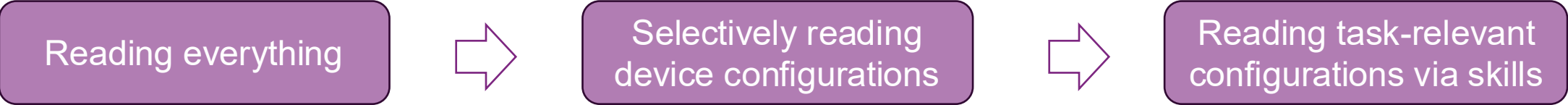
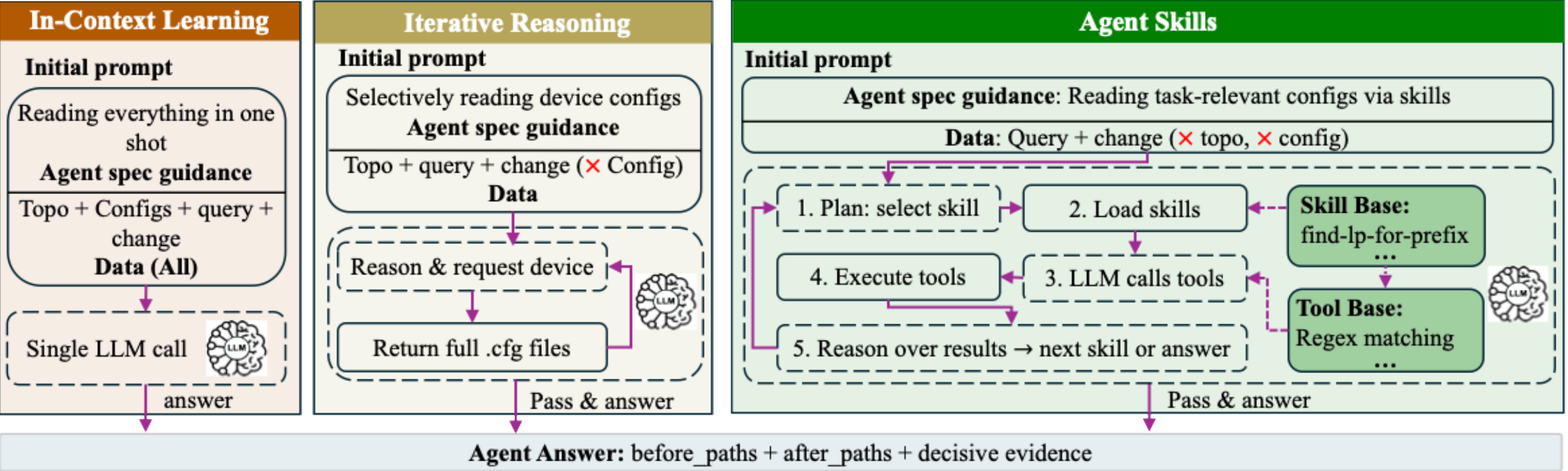
Routing Config Types
BGP Local Preference (LP)
Multi-Exit Discriminator (MED)
Static Route
BGP network advertisement
Prefix aggregation
Interface shutdown

Evaluated Agentic Paradigms

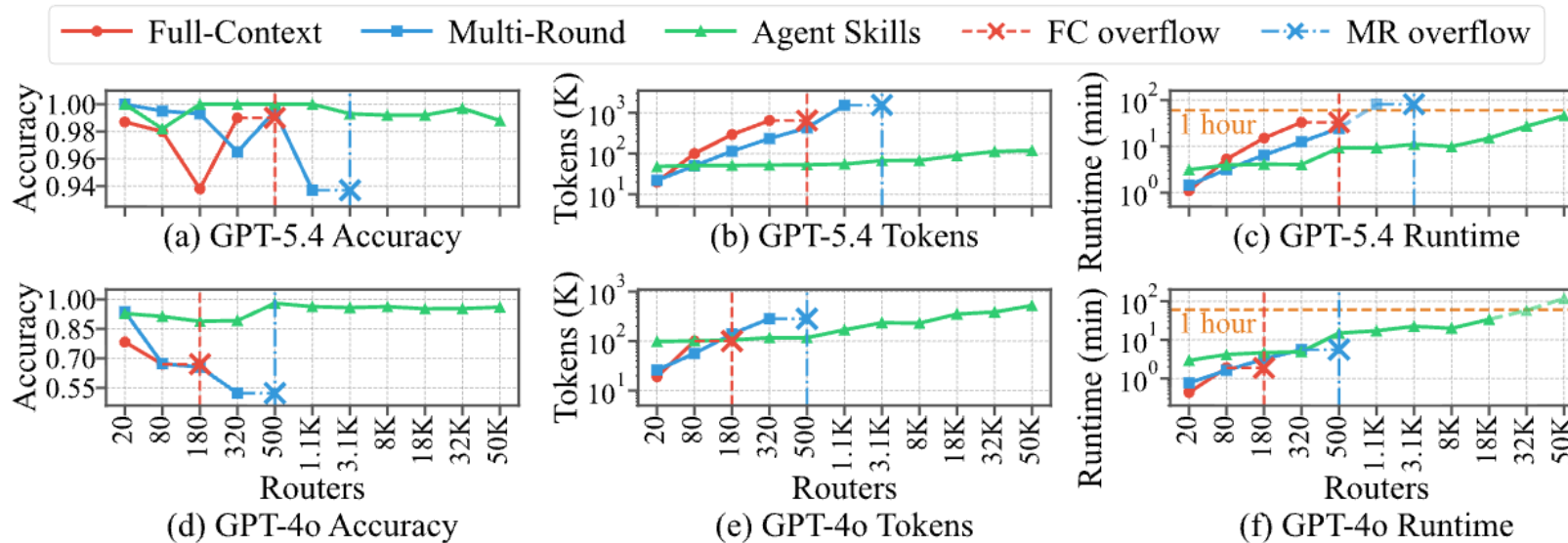
Three agentic paradigms in a unified framework. They follow a clear config-evidence exploration axis:



Shared guidance prompt (in each agent's initial prompt): Task description + CoT instruction + output schema + few-shot examples



Where Agents Shine: Scaling Path Analysis to 50K Routers



Across three agentic methods:

Agent Skills: highest accuracy, stable across scales.

Agent Skills: most stable in tokens & runtime.

Across different LLMs:

GPT-5.4 (Agent Skills) uses the fewest tokens and scales the farthest — beating thinking models (DeepSeek, o3).

Intra-DC LLM Comparison								
Rank	Model	k*	Acc	Tokens	t _{LLM}	t _{Runtime}	R _{LLM}	R _{Tool}
1	GPT-5.4	200 (50K)	99.5%	70K	212 s	778 s	7.8	29.5
2	Qwen-3.5-397B	120 (18K) [†]	99.3%	231K	351 s	1101 s	18.9	22.6
3	DeepSeek-V4-Pro	160 (32K) [†]	99.1%	178K	543 s	1331 s	15.8	29.2
4	OpenAI o3	160 (32K) [†]	96.8%	208K	323 s	1247 s	18.0	12.0
5	GPT-4o	120 (18K) [†]	93.8%	169K	172 s	830 s	15.7	15.9
6	Llama-3.3-70B	160 (32K) [†]	73.3%	408K	752 s	1910 s	23.5	24.3

Design principle — *Explore more; digest less.* Use skills to fetch sparse, decisive config evidence — don't read everything, don't over-think.

Where Agents Stumble: Cross-DC & Compound Changes

Inter-DC (GPT-5.4)							Compound Changes (GPT-5.4)						
k	Routers	Acc	Tokens	t _{LLM}	t _{Runtime}	R _{LLM}	k	Routers	Acc	Tokens	t _{LLM}	t _{Runtime}	R _{LLM}
4	48	96.2%	91K	276 s	329 s	7.6	4	20	99.5%	48K	147 s	178 s	6.4
16	672	90.0%	100K	305 s	418 s	8.0	16	320	96.2%	58K	177 s	245 s	6.4
50	6,350	83.7%	99K	302 s	921 s	7.9	50	3,125	80.5%	61K	177 s	571 s	6.4
200	100,400	73.5%	127K	387 s	3012 s	9.2	200	50,000	79.0%	58K	176 s	2100 s	6.3

Agent Skills (GPT-5.4) accuracy drops with scale in complex cases.

 **Lazy reasoning** — on long paths, the agent skips hop-by-hop analysis and falls back to the default next-hop (lowest router-ID).

Error pattern: inter-DC

 **Protocol entanglement** — when multiple protocols change the same flow, the agent fails to reason about cross-effects and often only analyzes one change.

Error pattern: compound changes

Ongoing Work

Push agent skills to production use.

RoutingBench will be open-sourced upon paper acceptance.

Seeking more insights on agentic config analysis.

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